Data Quality Assessment

Week 3 • Healthcare Operations & Financial Intelligence Dashboard • Track A: Data Analytics

Files reviewed

I inspected the five files below before building anything in Power BI. The encounters file carries most of the analysis, so it got the closest look.

File	Records	What I found
encounters.csv	27,891 rows, 14 columns	No duplicate IDs. START and STOP arrived as text. REASONCODE and REASONDESCRIPTION are blank on 19,541 rows.
patients.csv	973 rows	No duplicate IDs.
procedures.csv	47,701 rows	Used for procedure frequency and cost.
payers.csv	10 rows	Small lookup table of payers.
organizations.csv	1 row, 8 columns	Fully complete. It holds one organisation only, so I cannot compare organisations.

Issues found and what I did about them

1. Dates stored as text

START and STOP were imported as text strings, which cannot be subtracted to get a duration. I converted both to Date/Time in Power Query and calculated each encounter's duration as STOP minus START. This is what makes it possible to count encounters that lasted longer than 24 hours.

2. Missing reason fields

REASONCODE and REASONDESCRIPTION are empty on 19,541 of 27,891 encounters, which is about 70%. I did not delete these rows. They appear to be routine visits that were not logged with a specific reason or billing code. This is my reading of the data and not something the files state directly. Removing them would have thrown away most of the encounters and distorted every volume and cost figure.

Instead, I replaced the blanks in REASONDESCRIPTION with **No Reason Recorded**, so these visits show up as a named group in the dashboard and not as an empty label. REASONCODE was left as it came.

3. Duplicates

Neither encounters.csv nor patients.csv has a duplicate ID, so I removed no rows from them.

4. An incomplete year

Encounter counts jump around: 3,885 in 2014 and 3,530 in 2021, but only 220 in 2022. A drop that sharp looks like a partial extract and not a real fall in activity. I kept 2022 in the data but I do not read it as a decline.

Reconciliation checks

I recalculated the main figures from the reported counts and totals to make sure they agree.

Check	Reported	Recalculated
Ambulatory share: 12,537 of 27,891 encounters	44.95%	44.95%, matches
Long-stay share: 58 of 27,891 encounters	0.21%	0.21%, matches
Missing reason share: 19,541 of 27,891 rows	19,541 rows	70.1% of encounters
Payer coverage plus patient responsibility	$101.51M claims	$101.52M, within $0.01M of rounding

Limitations that affect the analysis

- Costs are recorded claim amounts. They do not show what was actually collected, and they exclude indirect overhead.
- Trend charts that include 2022 will understate that year.
- Because most reason fields are blank, I cannot break encounters down by reason with any confidence.
- The data comes from one organisation, so the findings describe that hospital only.